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Flood Prediction in Bangladesh Using Machine Learning and Explainable AI: A Comparative Study

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***Abstract*—Floods are among the most devastating natural disasters, posing significant threats to life, property, and economic stability, particularly in developing countries like Bangladesh. Accurate flood prediction remains a critical challenge, requiring the integration of advanced data analysis and machine learning techniques. This paper presents a comprehensive study using a dataset tailored specifically for Bangladesh to forecast floods. Five machine learning models—Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and Long Short-Term Memory (LSTM) networks—were employed to predict flood occurrences. Among these, the Gradient Boosting model achieved the highest accuracy of 95.38%. To enhance the interpretability of the model's predictions, Explainable AI techniques, specifically SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), were applied, providing insights into the model's decision-making process. This work not only contributes to the field of flood prediction in Bangladesh but also offers valuable information for policymakers to implement more effective flood management strategies.**

***Keywords*—Flood Prediction, Explainable AI(XAI), Bangladesh Floods, Disaster Management, Data-Driven Decision Making, AI in Disaster Preparedness.**

I. Introduction

When it comes to natural calamities, floods are among the most common and devastating. Areas that are low and have inadequate drainage are more likely to get flooded. Water causes floods when it rapidly covers typically dry ground. When a body of water like a lake, river, or sea overflows its banks, it can create flooding in neighboring countries. Lake and river levels change dramatically throughout the year as precipitation and evaporation patterns shift with the seasons. The location of rainfall also plays a role in the likelihood of flooding. Most floods may be predicted hours or days in advance, giving residents plenty of time to get out of harm's way. The economic and social environment has been severely disrupted by the floods. The global breakout of the coronavirus epidemic in 2020 exacerbated its consequences, wreaking havoc on economics and health around the world. Approximately 4.84 million Indians, 3.84 million Bangladeshis, and 3.28 million Chinese are affected by flooding annually, according to a poll [1]. In addition to low-lying countries like the Netherlands, Monaco, Bahrain, etc., areas with an elevation of less than 10 meters above sea level are at risk of flooding. From 1997 to 2008, floods in Australia claimed the lives of at least 73 individuals [2]. More than 147 million people will be impacted by floods by 2030, according to the World Resources Institute [3]. Annual property damage in urban areas is estimated to cost between 174 billion and 712 billion USD [3]. Because of its location in the world's lowest latitudes, Bangladesh is especially susceptible to natural disasters because eighty percent of the country is floodplain [4]. The destruction that can be caused by floods is enormous. When rivers flood or the sea moves inland, it can cause significant damage to many buildings. More than a million homes were destroyed by flooding in Bangladesh in July 2007 [5]. In reality, the damage caused by retreating flood waters might be significantly worse, including cracks in the framework, sharp objects, insecticides, gasoline, etc. There is a risk of water contamination due to untreated sewage. An increase in the prevalence of potentially fatal waterborne diseases like cholera, typhoid, and malaria. The primary goal of this study is to develop more precise methods for predicting floods. A flood data collection can be classified using binary classification if it contains labeled training data. It is possible to determine which model parameters are required. By fitting the model to the training dataset, it can be used to predict the labels of unseen data. Traditional ML methods are currently more accurate. This paper tries to improve the accuracy with which floods may be predicted using the dataset and several ML models.

II. Literature Review

Machine learning is commonly used in the field of flood prediction for data identification and classification, as well as in the classification and prediction of floods. Since floods are so common and so catastrophic, some people have tried to develop methods to forecast them. Some machine learning models have been developed in the past to forecast floods early to reduce the damage. Models to manage risks have also been researched. Recent research in India by S. Sankaranarayanan et al. [6] indicates that a variety of machine learning methods are employed to foretell floods. Observe and account for conditions including rainfall, humidity, temperature, and stream velocity. In this research, they evaluated the performance of various machine learning algorithms for flood prediction, focusing on Deep Neural Networks (DNN) and also considering Support Vector Machines, K-Nearest Neighbors, and Naive

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Bayes. In this study, the authors state that they employed the confusion matrix to determine the optimal method of flood detection by using temperature and precipitation as factors. Accuracy rates of SVM, Naive Bayes, and KNN in comparison to DNN's accuracy rate reveals some interesting trends. F. A. Ruslan et al. [7] offered a NNARX-structured artificial neural network model for flood prediction. The water levels of many upstream rivers are used as an input variable in this work because of their significance in flood formation. In this case, the model was trained first, and then it was used to verify the remaining samples. Use a simulation to foretell when a flood will occur. This model included four independent variables: upstream river water level, downstream river water level, flood site water level differential. Afterwards, they separated the information into a usable training set and a separate set of test data that the NNARX model requires as input. Flood levels were predicted using an artificial neural network technique called Back Propagation Neural Network in another study by R. Adnan et al. [8]. His BPN model uses readings from water gauges in the Malaysian state of Johor. The addition of the Extended Kalman Filter helped the model provide more desirable results (EKF). In this study, it was shown that including EKF into the BPN model enhances its predictive power. The predicting accuracy of SVM and ANN is compared in a publication by S.-Y. Liong et al. [9], which makes use of datasets of flood occurrences in Bangladesh. From what we can tell from this research, SVM can achieve prediction accuracy that is on par with, and even better than, ANN. As a result, it excels in performance compared to an ANN. The SVM was trained and validated with water level data from five stations in Bangladesh.

Although flood prediction using machine learning models have shown encouraging results, current research shows the significance of interpretability. In this perspective, explainable artificial intelligence (XAI) is becoming more popular in recent studies.

Biswajeet et al. [10] first presented a XAI model for mapping geographical flood susceptibility by combining convolutional neural network (CNN), deep learning models with the Shapley additive explanation (SHAP) model. Their research, carried out in South Korea's Jinju Province, produced a prediction accuracy of 88.4% according to the area under the receiver operating characteristic curve (AUROC). Numerous soil properties and land use greatly influenced the study area's flood susceptibility. The model outputs were made possible by the use of SHAP plots. This method not only produced precise forecasts but also found information about other variables that can affect flood susceptibility. This also enhances model's confidence and decision-making procedures.

Mahir et al. [11] also used XAI model in a study that was focused on Bangladesh, more specifically, the Sylhet region, to analyze water dynamics and forecast floods. Sylhet is prone to flooding. The study combined SHAP with deep learning models to research the connection between rainfall patterns in Northeastern India, the neighbouring upstream region, and floods in Sylhet. The study used Feedforward Neural Networks (FNN) and Long Short-Term Memory Recurrent Neural Networks (LSTM-RNN) to analyze time-series data related to flood preparedness. Compared to LSTM-RNN which showed significant temporal sensitivity with an R-squared of 0.94, FNN displayed more precision with an improved R-squared of 0.96. FNN also has a larger Root Mean Squared Error of 0.112, and a smaller Mean Squared Error of 0.02. Furthermore, this comparison method shed light on the underlying causes of flood occurrences in the area.

These studies show how explainable AI is becoming more and more popular for flood prediction. Through the integration of sophisticated machine learning models with interpretability methodologies, we can enhance flood preparedness and mitigate the damage further.

III. METHODOLOGY

A. Dataset Description

The dataset used in this study is based on 65 years of weather data from Bangladesh, originally sourced from the Bangladesh Meteorological Department (BMD) and preprocessed for public use on Kaggle [12]. This weather data, covering the period from 1948 to 2013, includes essential meteorological attributes such as Rainfall, Cloud Coverage, Relative Humidity, Minimum Temperature, Wind Speed, among others, and spans across 32 districts of Bangladesh, resulting in a total of 20,544 instances.This weather data was further enhanced in a previous study [13] by merging it with additional flood occurrence data for specific months and years, which had been gathered from various sources, including annual flood reports, newspapers, and research papers. The combined dataset, which serves as the basis for this research, was made available on GitHub. [14]

B. Dataset Preprocessing and Analysis

The dataset used in this study was largely prepared and ready for analysis. The primary preprocessing task involved modifying the "Flood?" column, which initially contained a value of 1 to indicate the occurrence of a flood, with no values for non-flood instances. To streamline the dataset for modeling purposes, we inserted a value of 0 in all instances where no flood occurred. This simple modification ensured the dataset was fully prepared for further analysis.

For the analysis, we conducted an area-wise flood mapping across Bangladesh to identify regions with higher flood percentages. This analysis allowed for a visual representation of the most flood-prone areas, which is crucial for targeted flood prediction and management efforts. The results of this analysis are illustrated in Figure 1, which highlights the flood distribution across different regions of Bangladesh.

Additionally, we performed a temporal analysis using a heatmap to identify which months are more prone to floods. This heatmap, covering the period from 1948 to 2013, reveals the flood occurrence patterns across different months, providing insights into the seasonal trends of flooding in Bangladesh. The heatmap, shown in Figure 2, clearly indicates the months

with the highest flood percentages, offering valuable information for understanding flood seasonality in the region.

C. Machine Learning Classifiers

In this paper, five different classifier algorithms were used to classify and predict floods.These are Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and Long Short-Term Memory (LSTM) networks

1) Logistic Regression: As with any other type of regression analysis, the results of a logistic regression can be used to make predictions. The analysis describes data and provides an explanation for the connection between a dependent binary variable and one or more independent variables. In most cases, a 1 indicates that something has occurred, while a 0 indicates that nothing has. The generalized linear model with a dichotomous dependent variable can be seen as the general framework for logistic regression. The Sigmoid function of the logistic regression model [15] is used for forecasting any dependent variable with just two possible values.

2) Random Forest: Random Forest is a supervised learning technique that may be applied in both classification and regression tasks. Yet, its primary application is in the realm of classification issues. For obvious reasons, the more trees there are in a forest, the more robust the forest will be. Like decision trees, random forest algorithms collect data samples, make predictions from each, and then vote on the best option. This is an ensemble method that improves upon decision trees individually by mitigating overfitting. This is the basic procedure for the Random Forest algorithm:

- Selecting random samples from a given dataset is the first step.
- The program then builds a decision tree for each sample to use.
- Next, it will collect the outcome of each decision tree's forecast.
- Each expected outcome is put to a vote.
- In the end, go with the most popular guess as your final guess.

3) Gradient Boosting: Gradient Boosting is an ensemble machine-learning technique that builds a powerful predictive model by combining several weak models, such as decision trees. To fix the mistakes that happened in the current ensemble model, Gradient Boosting iteratively adds new models [17]. After that, each successive model in the sequence to minimize the loss function and enhance the ensemble as a whole. For tasks involving regression and classification, such as the one we worked on, gradient boosting works well. In addition, it is tolerant to outliers and capable of handling sophisticated non-linear relationships in data. If it is not calibrated correctly, it may lead to overfitting. Additionally, the approach may be computationally demanding for very big datasets due to its sequential structure.

4) Xtreme Gradient Boosting, or XGBoost: XGBoost is a scalable ensemble technique that has demonstrated to be a reliable and efficient machine learning problem solver. [16] XGBoost enhances the Gradient Boosting model and makes it extremely effective. XGBoost uses a more regularized model formalization to prevent overfitting. This improves the performance as well. XGBoost is optimized for speed and efficiency, especially when tabular or structured data is being used. It uses methods like regularization, tree pruning, and parallel processing to improve accuracy and reduce computational stress. XGBoost can automatically determine the appropriate imputation and it can also manage missing variables internally. In recent times, XGBoost has gained much popularity for its high predictive capability.

5) Long Short-Term Memory (LSTM): Long Short-Term Memory (LSTM) is a specialized type of recurrent neural network (RNN) architecture designed to handle long-term dependencies in sequence data. Unlike standard feed-forward neural networks, LSTM has feedback connections and maintains an internal cell state. This allows it to remember information for long periods making LSTM particularly effective for time series forecasting, natural language processing, and other sequential data tasks. The LSTM architecture consists of a cell, an input gate, an output gate, and a forget gate. These gates regulate the flow of information into and out of the cell, enabling the network to learn which data in a sequence is important to keep and which is not. It also has the ability to handle the exploding / vanishing gradient problem. [18] By selectively updating or resetting its memory, LSTM can capture long-range dependencies that simple RNNs often struggle with, making it well-suited for tasks involving time series data like flood prediction.

IV. RESULT ANALYSIS

In this study, we evaluated the performance of five machine learning models: Logistic Regression, Random Forest, Gradient Boosting, XGBoost, and Long Short-Term Memory (LSTM) networks. The primary objective was to identify the model that provides the most accurate and reliable flood predictions for Bangladesh.

As illustrated in Table I, the Gradient Boosting model achieved the highest accuracy of 95.38%, outperforming all other models in terms of Accuracy, Precision, Recall, and F1 score, each reaching a value of 0.95. Gradient Boosting outperforms marginally due to its sequential approach to reducing residual errors, which allows for fine-grained optimization compared to the parallel tree-building methods of Random Forest and XGBoost. The trade-offs include Gradient Boosting's slightly higher computational cost versus XGBoost's speed and Random Forest's robustness to noisy data. This accuracy can also happen because of overfitting. To address overfitting, k-fold cross-validation was used to evaluate model performance, and results showed consistent generalization across folds. Additionally, the Gradient Boosting model's hyperparameters were tuned to balance bias and variance. Training and validation loss trends confirmed convergence without significant overfitting.

The Random Forest and XGBoost models also demonstrated strong performance, with accuracy scores of 94.92% and 94.47%, respectively. Both models exhibited a precision and

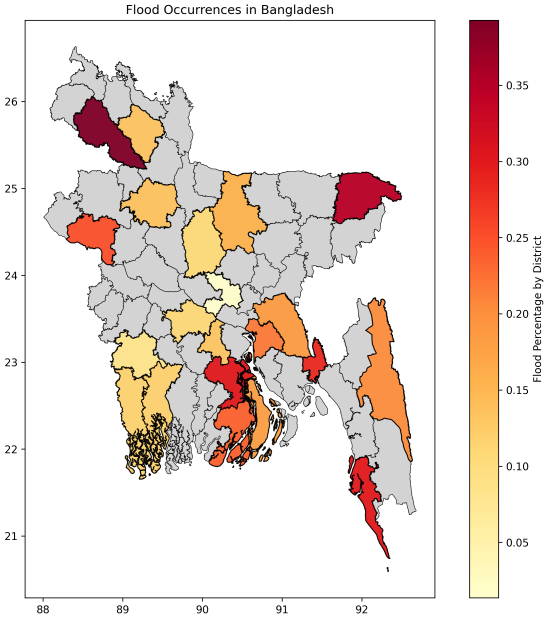


Fig. 1: Flood Percentage Map by Region

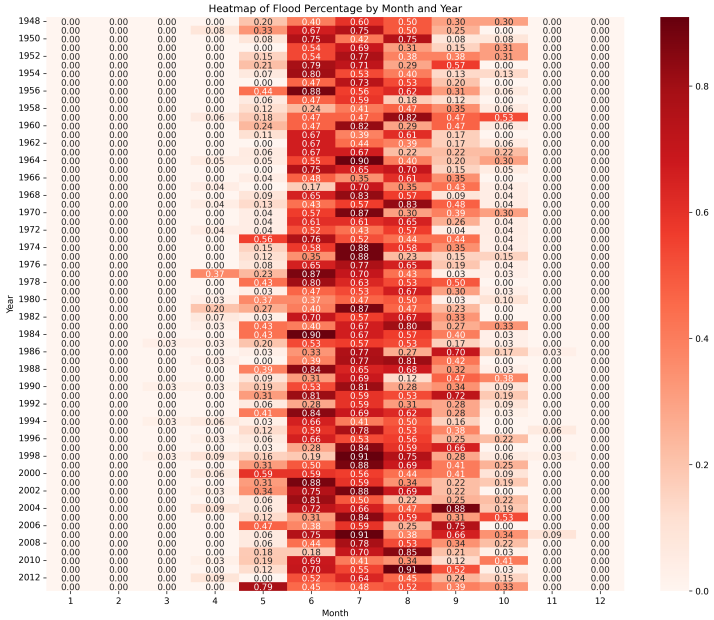


Fig. 2: Year and Month Wise Flood Percentage Heatmap

recall of 0.94, resulting in an F1 score of 0.94, which signifies a high level of consistency in predicting flood events.

Logistic Regression showed a slightly lower accuracy of 94.35%, along with precision, recall, and F1 scores of 0.94. Although its performance is robust, it does not match the precision of the ensemble methods.

The LSTM model, while effective in various time-series forecasting tasks, achieved the lowest performance in this context, with an accuracy of 85.80% and corresponding precision, recall, and F1 scores of 0.85. The LSTM model's underperformance likely stems from limited sequential patterns in the dataset, which primarily features aggregated monthly data rather than continuous time-series sequences. Additionally, tuning the model's hyperparameters, such as sequence length and learning rate, may have further limited its potential.

TABLE I: Comparison of Different Model's Performance

Model	Accuracy%	Precision	Recall	F1
Logistic Regression	94.35	0.94	0.94	0.94
Random Forest	94.92	0.95	0.95	0.95
Gradient boosting	**95.38**	**0.95**	**0.95**	**0.95**
XGBoost	94.47	0.94	0.94	0.94
LSTM	85.80	0.85	0.85	0.85

V. EXPLAINABILITY ANALYSIS

In order to understand the predictions made by our Gradient Boosting model, we conducted an explainability analysis using SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations). These methods provide insights into how different features contribute to the model's predictions, helping to explain why certain predictions were made.

1) SHAP Analysis: We analyzed four specific cases to demonstrate the model's decision-making process:

True Positive (TP): In this case, the model correctly predicted a flood event. As shown in the SHAP plot (Figure 3), features such as high Rainfall (feature Value of 1163) played a significant role in pushing the prediction towards a flood. There also a slight role of Bright Sunshine (feature Value of 3.5) for the prediction towards flood.

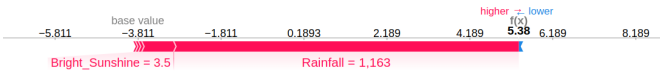


Fig. 3: True Positive with SHAP

True Negative (TN): This case represents a correct prediction where the model identified the absence of a flood. The SHAP plot (Figure 4) indicates that the most significant factor contributing to the prediction of flooding is rainfall, with a substantial negative SHAP value of -7.34. This suggests that low rainfall(feature Value of 43) significantly decreases the likelihood of flooding. Other features, such as bright sunshine (feature value of 7.8), minimum temperature (feature value of 15.7), relative humidity (feature value of 76), cloud coverage (feature value of 1.5), and maximum temperature (feature value of 33.1), also played roles in pushing the prediction towards a lower probability of flooding. However, their impacts were less pronounced compared to rainfall.

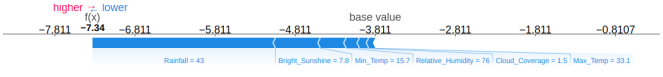


Fig. 4: True Negative with SHAP

False Positive (FP): In this instance, the model incorrectly predicted a flood. The SHAP plot (Figure 5) highlights the features that led to this false alarm, with moderately high Rainfall(feature value of 316), maximum temperature (feature value of 36.4), minimum temperature (feature value of 26.4), cloud coverage (feature value of 6.2), and bright sunshine (feature value of 4.2) features pushing the prediction over the threshold, despite the actual absence of a flood.

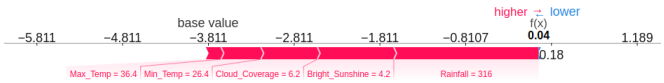


Fig. 5: False Positive with SHAP

False Negative (FN): Here, the model failed to predict a flood that did occur. The SHAP plot (Figure 6) shows that the most significant factor contributing to the prediction of false flooding is relative humidity, with a substantial negative fature value of 86. Other features, such as minimum temperature (feature value of 25.4), maximum temperature (feature value of 35.6), cloud coverage (feature value of 6.2), and rainfall (feature value of 204), also played roles in pushing the prediction towards a lower probability of flood prediction.

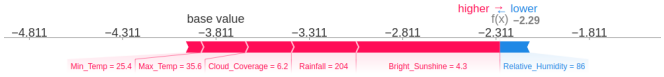


Fig. 6: False Negative with SHAP

2) LIME Analysis: LIME (Local Interpretable Model-Agnostic Explanations) breaks down the complex prediction of a machine learning model into simpler terms, focusing on the most important features that contributed to the outcome.

True Positive (TP): The LIME explanation reveals that model predicted a high probability of flooding (0.99) due to the combination of heavy rainfall (723.00 mm, significantly exceeding the threshold of 309.00 mm) and high cloud coverage (6.30, above the threshold of 5.50). While bright sunshine (3.90) was also considered, its impact was less significant compared to the other two factors.

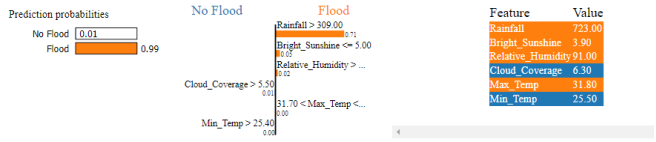


Fig. 7: True Positive with LIME

True Negative (TN): In this case the model predicted a low probability of flooding (0.00) due to the combination of low rainfall (43.00 mm, above the threshold of 8.00 mm), low cloud coverage (1.50, below the threshold of 1.60), and relatively high relative humidity (76.00, above the threshold of 75.00). While bright sunshine (7.80) was also considered, its impact was less significant compared to the other three factors.

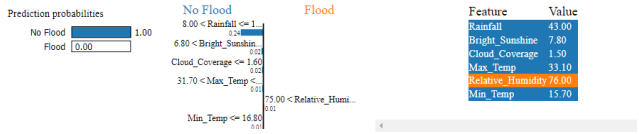


Fig. 8: True Negative with LIME

False Positive (FP): In this case the model predicted incorrectly a slight probability of flooding (0.51) due to the combination of high rainfall (316.00 mm, exceeding the threshold of 309.00 mm) and cloud coverage (6.20, above the threshold of 5.50). While bright sunshine (4.20) was also considered, its impact was less significant compared to the other two factors.

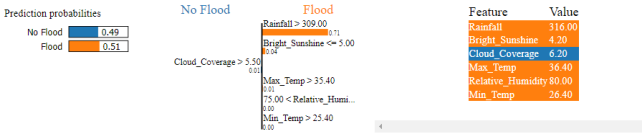


Fig. 9: False Positive with LIME

False Negative (FN): The LIME explanation reveals the model predicted wrongly of a low probability of flooding (0.09) due to the combination of rainfall (204.00 mm, above the threshold of 111.00 mm), low cloud coverage (6.20, above the threshold of 5.50), and relatively high relative humidity (86.00, above the threshold of 75.00). While bright sunshine (4.30) was also considered, its impact was less significant compared to the other three factors.

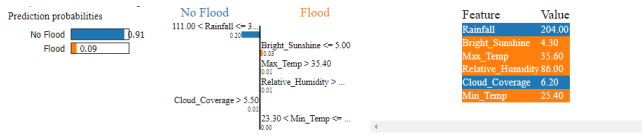


Fig. 10: False Negative with LIME

VI. Practical Implications of Explainability

The SHAP and LIME analyses provided critical insights into the model's decision-making process, which can significantly aid flood management strategies. Rainfall emerged as the most influential factor, followed by features like cloud coverage, relative humidity, and bright sunshine. These insights can enhance early warning systems by identifying thresholds for high-risk conditions and improving the accuracy of alerts. For instance, areas experiencing prolonged high rainfall combined with specific levels of relative humidity can be flagged for targeted interventions. Moreover, analyzing false positives and false negatives revealed patterns that can refine decision-making. For example, moderately high rainfall often led to false alarms, suggesting a need for region-specific adjustments in prediction thresholds. Similarly, underestimations in certain cases highlight the importance of reassessing the contribution of relative humidity and other secondary factors. These insights support better resource allocation, such as deploying

monitoring stations in vulnerable areas and improving infrastructure. By providing interpretable insights, the SHAP and LIME analyses bridge the gap between predictive modeling and practical flood management, supporting more targeted and effective mitigation strategies.

VII. Potential Approaches for Future Development

Future research could focus on several key areas to enhance the model's effectiveness. Incorporating additional data sources, such as real-time hydrological data, satellite imagery, and river water levels, could significantly improve predictive capabilities. Additionally, exploring advanced modeling techniques, including hybrid models and deep learning approaches tailored to temporal data, may further refine prediction accuracy. Developing region-specific models that account for the unique geographical and climatic conditions of different areas in Bangladesh could lead to more localized predictions. Moreover, the implementation of a real-time flood prediction system that continuously updates predictions based on incoming data would be valuable for timely warnings. Finally, integrating model predictions with disaster management policies could ensure that these predictions lead to actionable strategies for flood prevention and mitigation. These future directions offer a pathway to further improving flood prediction models and their practical applications in managing flood risks.

VIII. Conclusion

Floods are devastating natural calamities and even more severe for a country like Bangladesh. The country has suffered heavily because of floods in previous years. Floods occurring in 1988, 1998, 2004, 2007, 2017, and even the recent flood in Feni, Noakhali of 2024 have caused serious financial damage along with a heavy death toll. The recent flood in the northern region of the country has also left a grim mark. Considering all these, it is of utmost importance to manage floods effectively to reduce damage. Predicting floods accurately beforehand thus plays a significant role in this aspect.

The main purpose of this paper was to predict floods more accurately compared to the existing works, along with finding the reasoning behind the model's predictions. The proposed model has demonstrated promising results in predicting flood occurrences, outperforming existing approaches. The explainability techniques employed, such as SHAP and LIME, have provided valuable insights into the model's decision-making process, enhancing its transparency and trustworthiness. However, deployment challenges remain, including ensuring computational efficiency for real-time flood predictions in resource-constrained regions. Optimizing model architectures and leveraging cloud-based solutions can address scalability concerns. Furthermore, integrating real-time data streams from weather sensors and satellites would significantly enhance the model's applicability and reliability in dynamic environments.

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